Yue Pan

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Education

University College London (UCL) **Sep. 2023 – Jun. 2027**
MEng in Computer Science *London, United Kingdom*

Beijing National Day School **Sep. 2020 – Jun. 2023**
High School Diploma *Beijing, China*

Work Experience

ByteDance – Data-Arch / TikTok Push AI Infrastructure **Jun. 2026 – Sep. 2026**
AI Infrastructure Intern *Shanghai, China*

- Designed and developed a **general-purpose Agent framework** adopted internally by the TikTok Push team, supporting task planning, tool invocation, Skill execution, trajectory collection, and automated evaluation.
- Developed a trace to skill pipeline that transforms Agent trajectories into reusable and versioned Skills through trajectory structuring, experience abstraction, verification, and iterative refinement.
- Investigating a **Skill–Model** continual learning framework that applies **on-policy context-conditioned self-distillation (OPSD)** to distill verified Skills into model parameters, enabling Agent capabilities to evolve jointly through explicit Skills and implicit model knowledge.
- Built On-call Agent and Coding Agent applications on top of the framework for incident diagnosis and repository-level engineering tasks, serving as internal testbeds for evaluating continual learning and Skill evolution in real-world workflows.

UCL SOLAR Group **Mar. 2025 – Present**
Research Assistant *London, United Kingdom*

- Led the development of **Prometheus** under the supervision of Prof. He Ye, an automated software issue-resolution Agent integrating knowledge graphs with a multi-agent architecture.
- Achieved an overall **global Top-7** ranking on SWE-bench Verified and ranked **Top-1 among GPT-5-based systems**; the corresponding research paper has been submitted to ICSE 2027.
- Designed and implemented **Athena**, a long-term memory module that unifies semantic and episodic memory, significantly improving long-horizon context understanding and reasoning for coding Agents.

Peking University **Jun. 2025 – Sep. 2025**
Research Assistant *Beijing, China*

- Conducted research on context compression for coding Agent systems under the supervision of Prof. Sergey Mechtaev.
- Investigated methods for reducing model input length while preserving semantic fidelity, improving reasoning efficiency in software engineering tasks.
- Constructed a large-scale code-context dataset and fine-tuned models for efficient context summarization and compression.

Publications

CRJudgeBench: Can AI Detect Plausible but Invalid Code Reviews? **2026**
***Yue Pan**, Jiawei Li, Ziyuan Zhang, Xiangxin Zhao, He Ye* *Submitted to ICLR 2027*

- Introduced **CRJudgeBench**, a benchmark of **1,199 human-verified code-review instances** for evaluating whether review comments are technically correct in their repository context.
- Developed **Sentinel**, a repository-grounded judge Agent that actively gathers code evidence to verify review comments, trained through **iterative action-level learning from a privileged teacher**.
- Achieved **76.60% accuracy** on the 359-instance test set, outperforming the base model by **19.78 percentage points** and the strongest evaluated baseline by **6.13 percentage points**.
- Paper: https://arxiv.org/abs/2609.37216

Prometheus: Long-Horizon Issue Resolution with Knowledge Graphs and Agent Memory **2026**
***Yue Pan**, Zimin Chen, Siyu Lu, . . . , Claire Le Goues, Martin Monperrus, He Ye* *Submitted to ICSE 2027*

- Proposed **Prometheus**, an automated software issue-resolution Agent integrating knowledge graphs with a multi-agent architecture.
- Designed **Athena**, a long-term memory module that unifies semantic and episodic memory to improve long-horizon context understanding and reasoning for coding Agents.
- Achieved a global Top-7 overall ranking on SWE-bench Verified and ranked Top-1 among systems based on GPT-5-family models.
- Paper: https://arxiv.org/abs/2507.19942

CONTEXTBENCH: A Benchmark for Context Retrieval in Coding Agents **2026**
*Han Li, Letian Zhu, . . . , **Yue Pan**, Earl T. Barr, Federica Sarro, He Ye* *Submitted to ICLR 2027*

- Participated in **CONTEXTBENCH**, the first **process-oriented benchmark** for evaluating context retrieval in coding Agents.
- Constructed a dataset of 1,136 real-world repository-level tasks spanning 66 code repositories and eight programming languages, with manually annotated Gold Context.
- Conducted an empirical evaluation of leading coding Agents and frontier language models, revealing key limitations such as high recall but low precision and insufficient utilization of retrieved context.

Technical Skills

Programming Languages: Python, Java, JavaScript, C/C++, Rust, SQL, HTML/CSS
Frameworks and Libraries: Django, Spring Boot, FastAPI, Flask, React, PyTorch, TensorFlow
Systems and Tools: Linux, Docker, Kubernetes, Nginx, Redis
Databases: PostgreSQL, MySQL

Honors and Awards

AMC 12A Mathematics Competition **Nov. 2021**
Global Top 2.5% *Beijing, China*

Euclid Mathematics Contest **Apr. 2022**
Certificate of Distinction *Beijing, China*